

WATERTOWN, WI, USA

WMO: 726464

Lat: 43.167N

Lon: 88.717W

Elev: 820

StdP: 14.27

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 54834

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.4	-1.4	-15.1	2.5	-4.7	-9.5	3.4	0.7	22.5	22.5	20.1	25.0	4.7	280	0.520

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.3	89.7	75.3	86.3	73.8	83.7	72.6	79.2	84.9	76.8	83.0	74.8	80.9	9.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.4	146.9	82.3	74.9	134.8	80.3	72.8	125.5	78.2	43.4	85.2	40.7	82.6	38.9	81.6	88.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.6	18.3	16.4	-14.1	93.4	6.3	3.5	-18.6	95.9	-22.3	98.0	-25.8	99.9	-30.4	102.4		
			-14.6	82.5	6.1	2.7	-19.0	84.5	-22.6	86.1	-26.1	87.6	-30.5	89.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.3	20.1	23.4	34.3	46.4	57.8	67.5	71.6	69.7	62.7	50.4	37.2	25.4
	DBStd	20.33	11.88	12.06	11.64	9.44	8.27	6.73	5.70	5.37	7.90	9.14	10.00	10.87
	HDD50	3651	927	745	508	174	21	0	0	0	5	110	399	762
	HDD65	7058	1392	1164	954	560	255	49	9	17	135	462	835	1226
	CDD50	2677	0	1	20	68	261	526	670	611	385	121	13	1
	CDD65	613	0	0	2	4	31	125	214	163	65	9	0	0
	CDH74	5657	0	0	13	70	345	1199	2021	1308	606	94	1	0
	CDH80	1638	0	0	1	12	83	365	663	341	158	15	0	0
Wind	WSAvg	6.7	7.4	7.4	7.7	8.5	7.4	5.9	5.1	4.5	5.4	6.8	7.2	7.1
Precipitation	PrecAvg	33.5	1.3	1.1	2.0	3.0	3.3	4.3	4.2	4.0	3.8	2.5	2.3	1.6
	PrecMax	44.1	3.5	3.3	5.0	7.0	9.7	9.4	10.5	9.2	13.5	6.3	6.1	3.9
	PrecMin	19.6	0.1	0.0	0.2	0.8	0.7	1.2	0.7	0.6	0.4	0.2	0.2	0.2
	PrecStd	5.1	0.8	0.8	1.2	1.3	1.9	2.2	2.0	2.0	2.7	1.5	1.3	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.0	56.9	74.9	81.6	87.7	91.3	95.3	90.7	89.1	82.0	68.5	56.6
		MCWB	47.4	51.0	63.7	62.3	70.3	74.8	78.2	77.7	72.9	67.6	58.7	55.2
	2%	DB	43.1	48.1	64.1	74.6	81.9	87.8	89.9	87.3	84.2	75.8	62.0	47.8
		MCWB	39.4	43.8	56.5	59.7	67.4	74.3	76.4	75.6	72.0	64.9	54.5	44.3
	5%	DB	38.5	43.2	57.2	68.3	78.0	84.0	86.4	83.7	81.1	71.5	56.9	43.3
		MCWB	36.4	39.3	50.8	55.7	64.9	71.7	74.5	73.7	69.1	63.1	51.1	40.5
	10%	DB	36.1	38.7	51.8	63.2	73.1	81.3	83.6	81.2	77.2	65.8	53.4	39.1
		MCWB	34.7	36.2	46.1	52.5	62.0	70.2	73.6	71.6	67.0	59.5	48.4	36.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.9	52.3	65.5	66.6	74.6	80.6	82.8	81.8	77.3	72.0	61.2	55.0
		MCDB	47.3	54.8	71.6	76.2	82.4	86.7	87.9	86.8	82.9	78.4	65.5	56.7
	2%	WB	39.7	44.9	59.0	62.0	70.7	77.1	79.5	78.2	74.4	67.8	56.6	45.5
		MCDB	42.1	47.3	63.9	70.6	78.0	83.3	85.3	83.7	80.5	73.2	60.9	46.5
	5%	WB	36.8	39.9	52.5	58.1	67.8	74.4	77.1	75.5	71.9	64.3	52.3	40.9
		MCDB	38.2	42.7	56.7	66.8	74.9	81.3	83.9	80.8	78.4	68.8	56.1	43.2
	10%	WB	34.3	36.7	45.6	53.8	64.5	72.0	75.1	73.6	69.3	60.3	47.8	37.2
		MCDB	35.8	38.6	49.9	61.9	70.6	79.1	81.3	78.8	74.2	65.1	52.1	39.3
Mean Daily Temperature Range	5% DB	MDBR	14.3	15.6	17.4	20.1	20.9	20.2	20.3	19.6	21.9	19.2	15.8	13.4
		MCDBR	15.8	19.4	24.5	29.5	26.0	22.9	22.3	21.8	25.1	26.0	22.5	16.4
		MCWBR	13.9	15.1	16.8	16.9	14.6	12.3	11.9	12.3	13.6	15.8	16.5	13.6
	5% WB	MCDBR	14.6	17.9	22.3	25.6	22.4	20.4	19.6	18.2	21.2	21.7	21.2	14.9
		MCWBR	13.2	14.2	17.2	17.1	14.8	12.9	12.7	11.6	13.2	15.7	17.0	13.6
Clear-Sky Solar Irradiance	taub	0.279	0.315	0.336	0.382	0.403	0.422	0.419	0.406	0.375	0.337	0.298	0.277	
	taud	2.385	2.279	2.368	2.278	2.269	2.255	2.263	2.290	2.367	2.470	2.512	2.432	
	Ebn at Noon	273	278	287	280	276	270	270	269	270	269	266	264	
	Edh at Noon	24	33	34	40	42	43	42	39	34	27	22	22	
All-Sky Solar Radiation	RadAvg	521	849	1188	1451	1705	1913	1938	1674	1366	875	564	408	
	RadStd	72	92	130	128	130	136	93	105	113	101	48	38	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	-2.74	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page